

ANIMOVE



```
P1.x=diag(c(0, 0.001, 0.001))
P1.y=diag(c(0, 0.001, 0.001))

displayPar(mov.model=~1, err.model=list(x=~errX, y=~errY), drift.no=1,
  data=nfsNew, fixPar=c(NA, 1, NA, 1, NA, NA, NA, NA))

t <- crwMLE(mov.model=~1, err.model=list(x=~errX, y=~errY), drift.no=1,
  data=nfsNew, coord=c("longitude", "latitude"), polar.coord=0,
  Time.name="Time", initial.state=initial.drift,
  fixPar=c(NA, 1, NA, 1, NA, NA, NA, NA),
  control=list(maxit=2000, trace=1, REPORT=10),
```



December 2025

Trajectory Centered Analysis

Variance Components of Movement

Variance components of movement

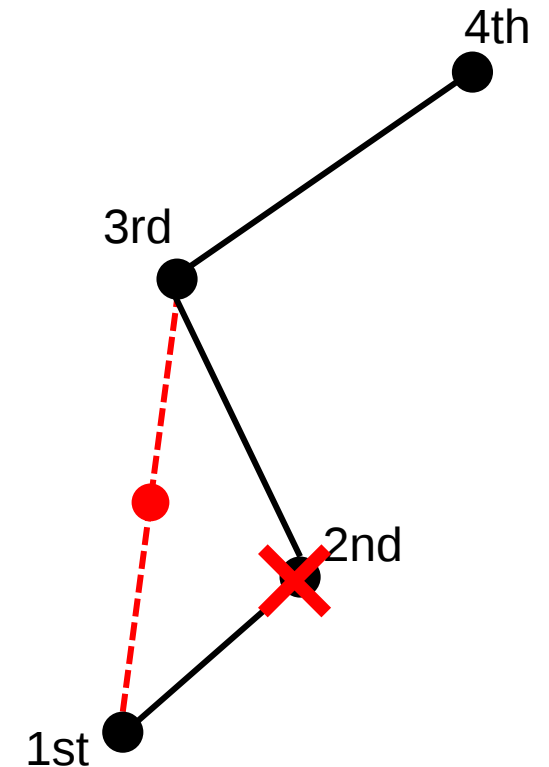
Variance estimation is based on a **leave-one-out method** where from 3 consecutive locations the 2nd one is omitted. Based on the timestamp of the 2nd location, a location is estimated on a straight line between location 1 and 3. **The distance between the actual and the estimated 2nd location represents the amount of variance.**

Can deal with irregular/gapy data.

Implementation in R package `move`.

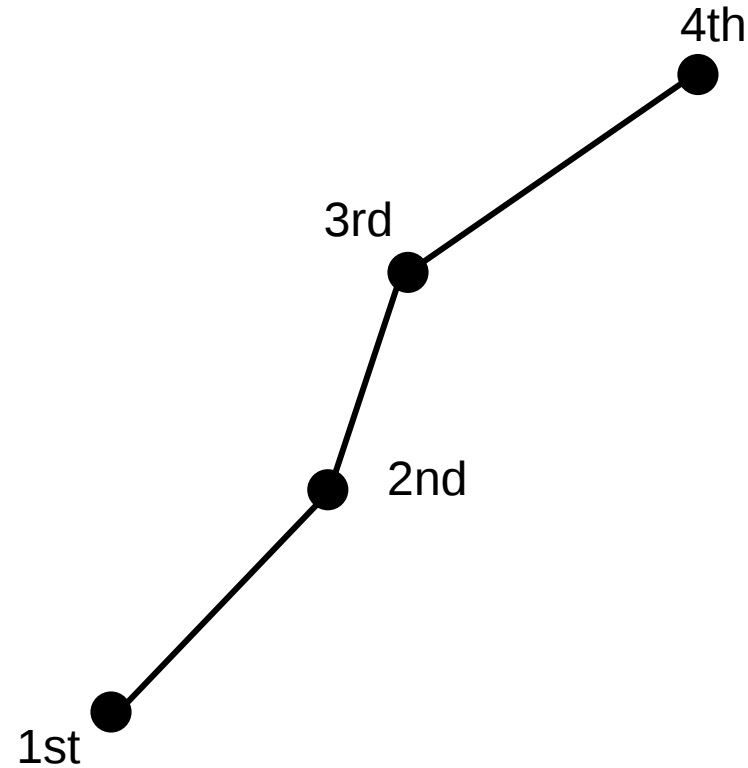
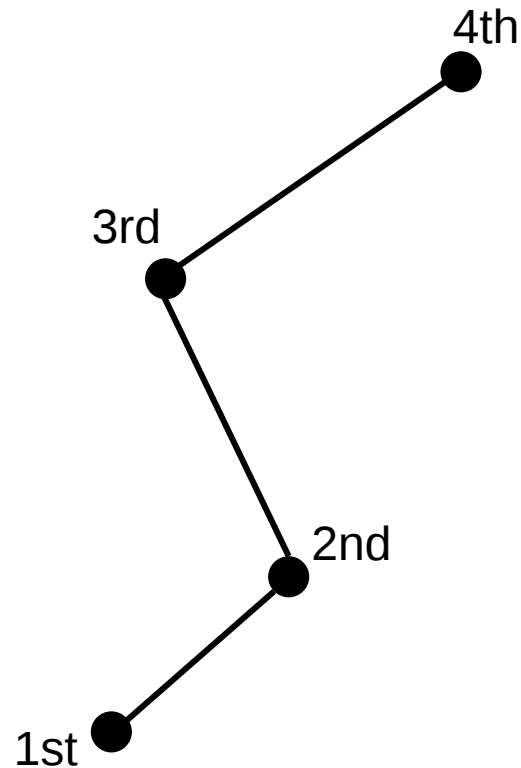
Estimated variances of the movement process are dynamic, **can change over a trajectory.**

- Variance of **dBB**: changes in speed and/or deviation from straight line
- Variances of **dBGB**: parallel (changes in speed) and orthogonal (deviation from the straight line)



Variance of dBB (Dynamic Brownian Bridge Movement Model)

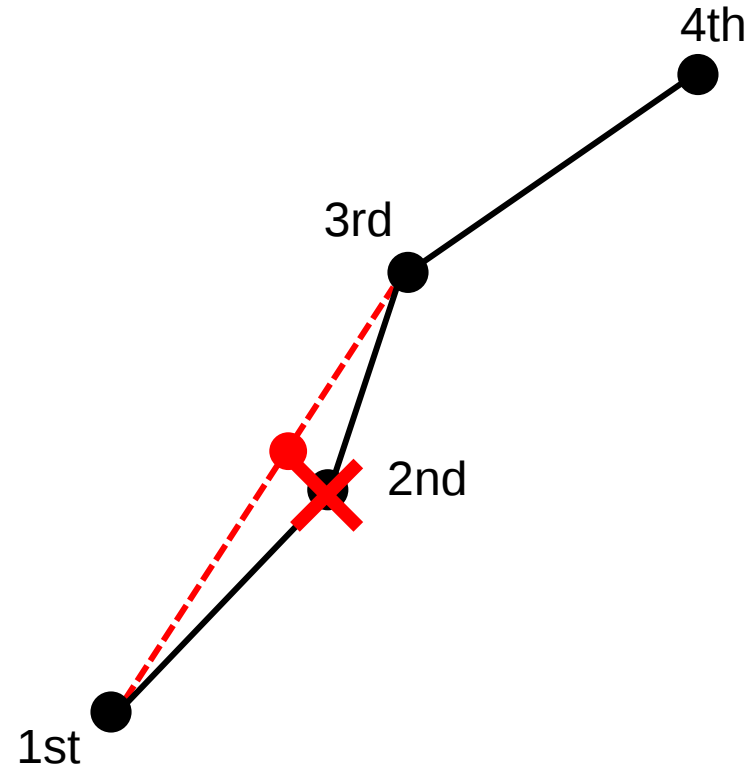
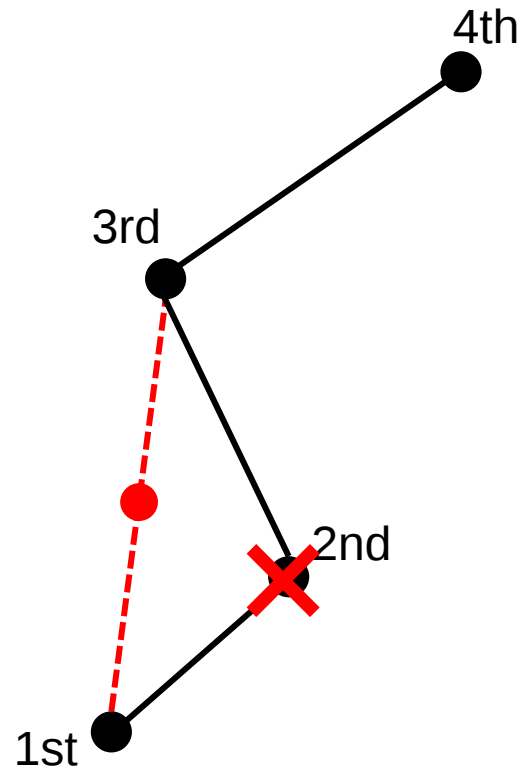
dBB allows to assigns a probability of **where the animal could have been when we did not observe it (between two known locations)**. It assumes brownian motion between consecutive points and returns the probability of being anywhere between any two points at any given time.



Variance of dBB (Dynamic Brownian Bridge Movement Model)

The estimated probability is based on the movement variance of the trajectory: **the faster and more direct, the less the variance** between the two points.

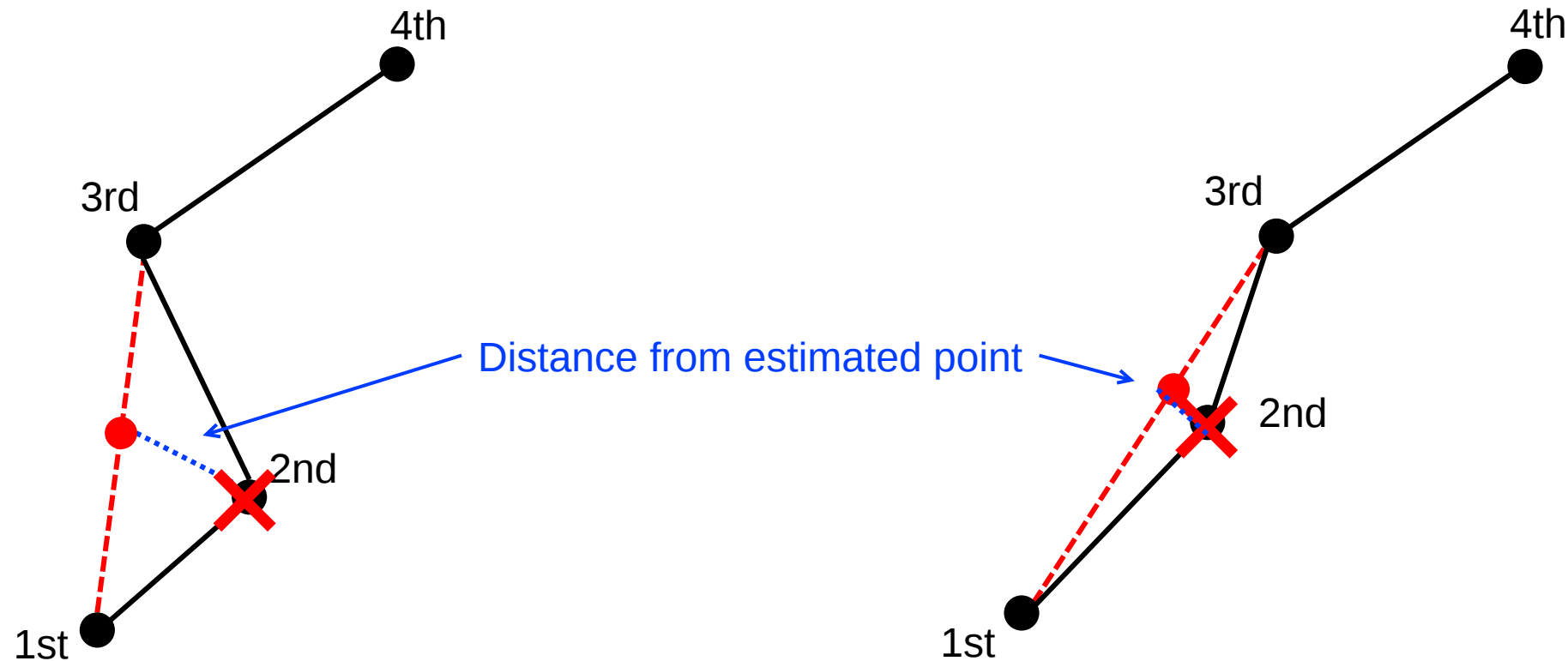
Variance is calculated leaving out the 2nd of three locations and calculating its **distance between the observed 2nd location and the estimated 2nd location**.



Variance of dBB (Dynamic Brownian Bridge Movement Model)

The estimated probability is based on the movement variance of the trajectory: **the faster and more direct, the less the variance** between the two points.

Variance is calculated leaving out the 2nd of three locations and calculating its **distance between the observed 2nd location and the estimated 2nd location**.



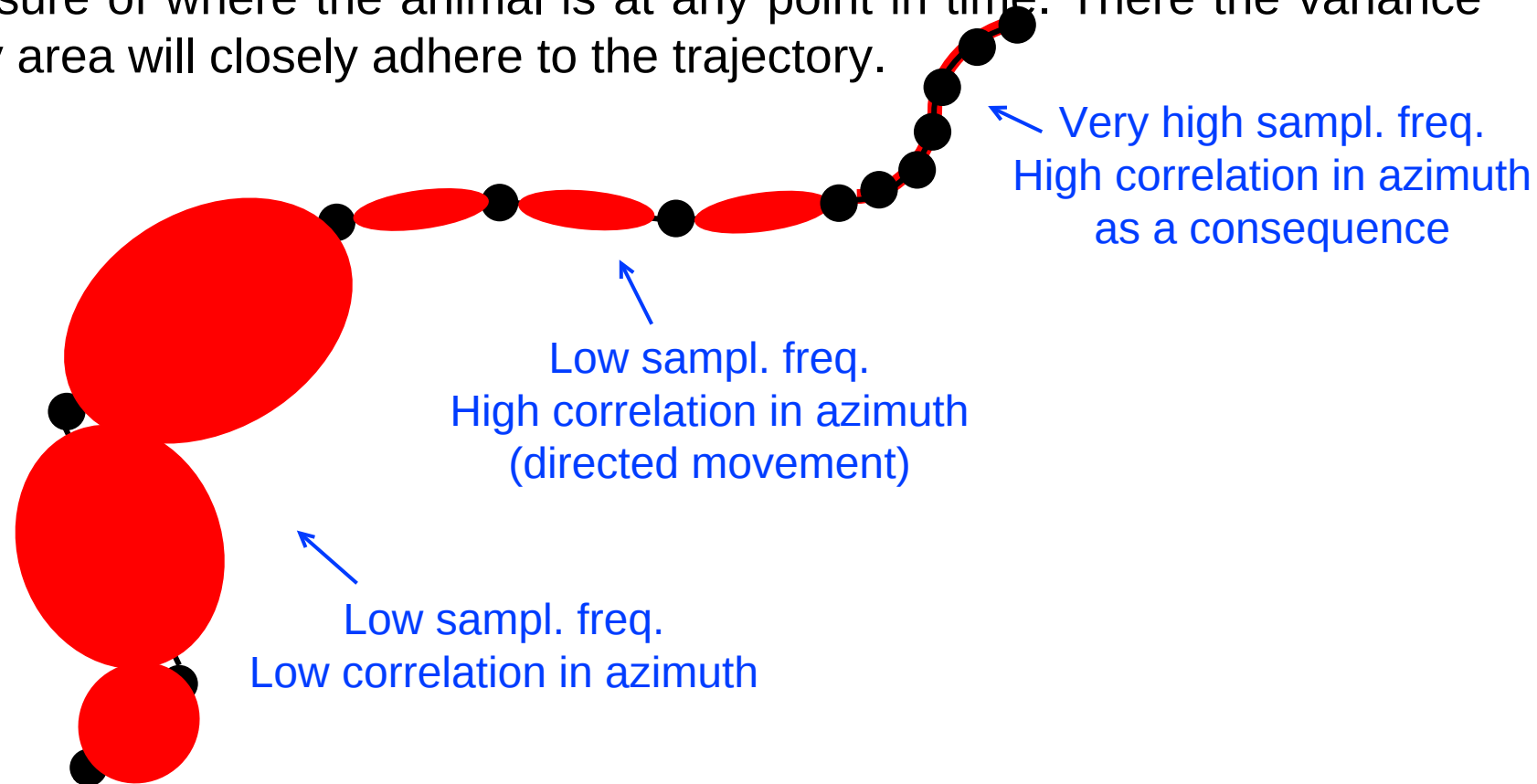
Variance of dBB (Dynamic Brownian Bridge Movement Model)

Based on the calculated **variance**, the dBB will be able to calculate, for each time gap between any two consecutive locations, **the probability of where the animal could have been**.

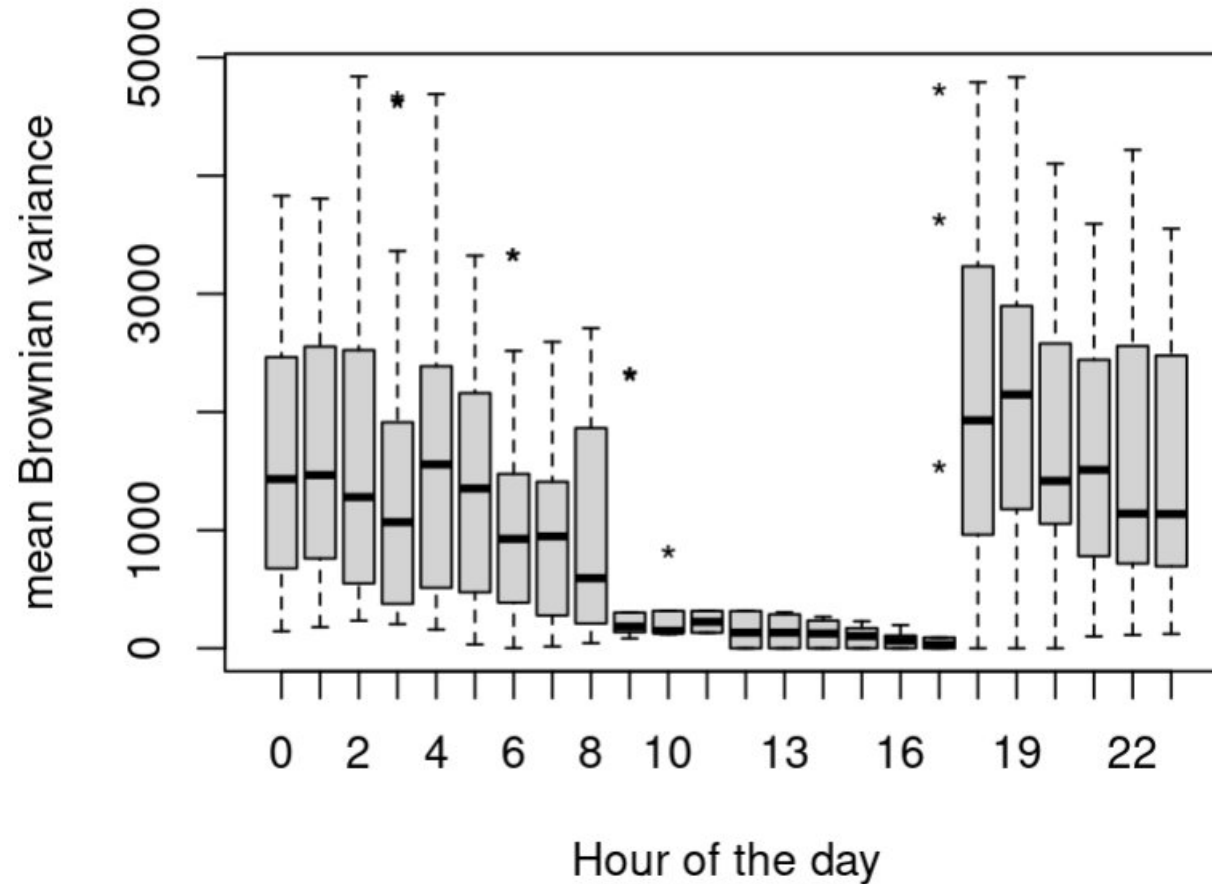
The smaller the time gap and the more consistent the movement process, the smaller the probability area between locations, which is allowed to vary along the trajectory.

With a 1 Hz sampling rate we are sure of where the animal is at any point in time. There the variance will be close to 0 and the probability area will closely adhere to the trajectory.

You will see dBB again tomorrow in relation to **space use**. The estimated probability can be used to calculate the area used by an animal during the time of tracking:
Utilization Distributions (occurrence ranges).



Variance of dBB (Dynamic Brownian Bridge Movement Model)

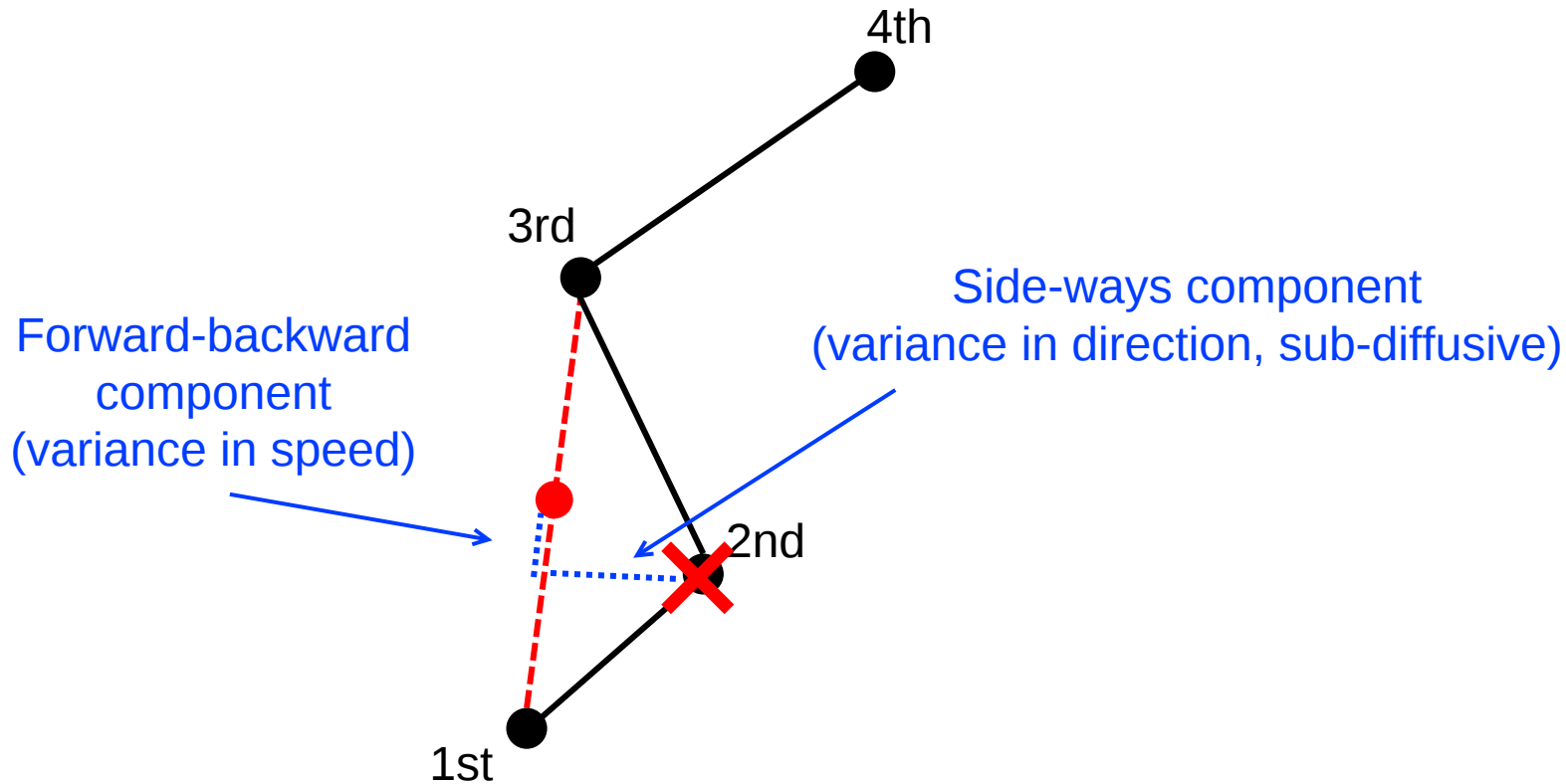


Leroy moves around during the night, and sleeps during the day. Low variance in the motion variance means **the location I left out is easy to predict (either stationary, or moving with constant speed).**

Variance of dBGB (Dynamic Bi-Gaussian Bridge Movement Model)

The variance in dBGB decomposes the variance in two:

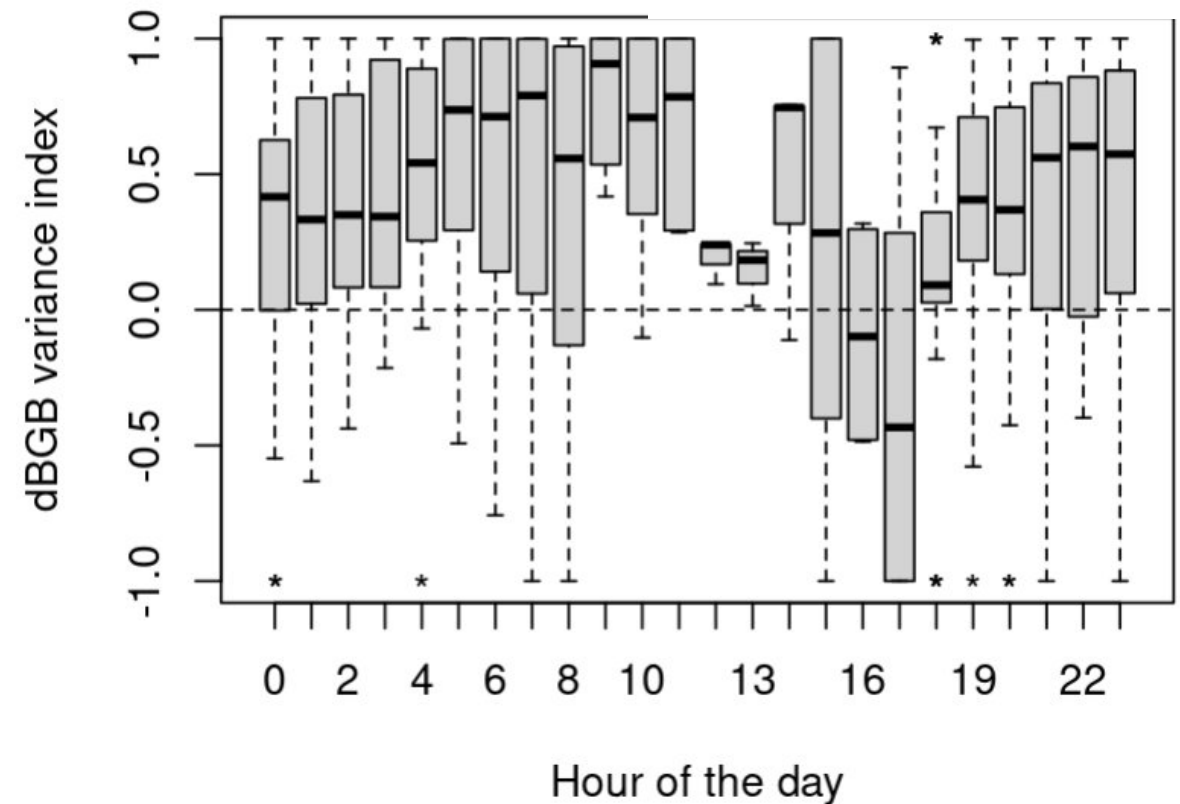
Forward-backward (parallel to movement) and **side-ways component** (perpendicular to movement). It helps distinguishing stationary from movement (which resulted in similar dB variance above).



Variance of dBGB (Dynamic Bi-Gaussian Bridge Movement Model)



- **Values around zero:** true Brownian motion, diffusive in all directions, as likely to go sideways than front or back. (Almost nothing in nature is brownian, but **GPS error** is, if data fall on 0 it could be error e.g. in the den, or the sampling frequency could be so low that autocorrelation is removed).
- **Positive values:** directional movement, more front or back than side, less changes in direction and more in velocity.
- **Negative values:** more likely to go sideways, there are more changes in direction and less in velocity, e.g. search behaviour.



Hands on

How to do this in R in the script:
“6C_TrajectoryAnalysis_dBBMM.R”